February 13, 2026

Recall:

- A point $\vec{a}$ in a set X inside of $\mathbb{R}^n$ (i.e. $\vec{a} \in X \subseteq \mathbb{R}^n$) is called $\boxed{\text{isolated point of } X}$ if there is $\delta > 0$ such that <u>no</u> points $\vec{x}$ in X satisfy $0 < \|\vec{x} - \vec{a}\| < \delta$ (i.e. $\{\vec{x} \in X : \|\vec{x} - \vec{a}\| < \delta\} = \{\vec{a}\}$, or $B_\delta(\vec{a}) \cap X = \{\vec{a}\}$).

- A function $f : X \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$ is $\boxed{\text{continuous at } \vec{a}}$ in X if either: $\vec{a}$ is isolated in X, or
$$\lim_{\vec{x} \to \vec{a}} f(\vec{x}) = f(\vec{a}).$$

- We say $f : X \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$ is $\boxed{\text{continuous}}$ if f is continuous at <u>every</u> $\vec{a}$ in X

Theorem: The following functions are continuous (and their domain has no isolated points):

1) Any polynomial $p : \mathbb{R}^n \to \mathbb{R}$ (e.g. $p(x, y, z) = x^2 y + z^3$)

2) Any rational function $f(\vec{x}) = \dfrac{p(\vec{x})}{q(\vec{x})}$
   (where $p, q : \mathbb{R}^n \to \mathbb{R}$ are polynomials)
   $\text{domain}(f) = \mathbb{R}^n \setminus \{\vec{x} \in \mathbb{R}^n : q(\vec{x}) = 0\}$

3) exponential $e^x : \mathbb{R} \to \mathbb{R}_{>0}$ and
   logarithm $\ln : \mathbb{R}_{>0} \to \mathbb{R}$

4) Trig functions, $\cos, \sin : \mathbb{R} \to [-1, 1]$
   $\tan, \sec, \csc, \cot$ (from their respective domains)

Theorem: $F : X \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is continuous at $\vec{a}$ in X if and only if each of its component function ($F_i : X \to \mathbb{R}$ for $i = 1, \dots, m$) is continuous at $\vec{a}$.

e.g. is $f(x,y) = (3x^2+y, \sin(x), e^x)$ continuous?

$f(x,y) = (f_1(x,y), f_2(x,y), f_3(x,y))$
$f_1(x,y)$ polynomial $\rightarrow$ cont. on all $\mathbb{R}^2$
$f_2(x,y)$ sine $\rightarrow$ cont on all $\mathbb{R}^2$
$f_3(x,y) = e^x$ exponential $\rightarrow$ cont. on all $\mathbb{R}^2$

e.g.: is $f(x) = \frac{1}{x}$ continuous?
(Remember f is continuous if it is continuous at every point of its domain)
domain $(f) = \mathbb{R} \setminus \{0\}$

this is continuous (on its domain)

e.g. is $f(x,y) = \frac{xy}{x^2+y^2}$ continuous?
(Remember f is continuous if it is continuous at every point of its domain)

this is continuous (on its domain $= \mathbb{R}^2 \setminus \{(0,0)\}$

e.g. is $f(x,y) = \begin{cases} 0 & , \text{if } (x,y) = (0,0) \\ \frac{xy}{x^2+y^2} & , \text{if } (x,y) \neq (0,0) \end{cases}$

why?

this is not continuous at $(0,0)$
(but it is continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$

$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ DNE we can show this taking limits along different paths

Continuity Laws :

Suppose :

- $F : X \subseteq \mathbb{R}^m \to \mathbb{R}^n$ is cont. at $\vec{a} \in X$
- $G : X \subseteq \mathbb{R}^m \to \mathbb{R}^n$ is cont. at $\vec{a} \in X$
- $f : X \subseteq \mathbb{R}^m \to \mathbb{R}$ is cont. at $\vec{a} \in X$
- $g : X \subseteq \mathbb{R}^m \to \mathbb{R}$ is cont. at $\vec{a} \in X$

then :

1) $F(\vec{x}) + G(\vec{x})$ is cont. at $\vec{a} \in X$
2) $f(\vec{x})\, F(\vec{x})$ is cont. at $\vec{a} \in X$
3) if $k \in \mathbb{R}$, then $k\,F(\vec{x})$ is cont. at $\vec{a} \in X$
4) $f(\vec{x})\, g(\vec{x})$ is cont. at $\vec{a} \in X$
5) $f(\vec{x})/g(\vec{x})$ is cont. at $\vec{a} \in X$ as long as $g(\vec{a}) \neq 0$ and $g(\vec{x}) \neq 0$ for $\vec{x}$ near $\vec{a}$.

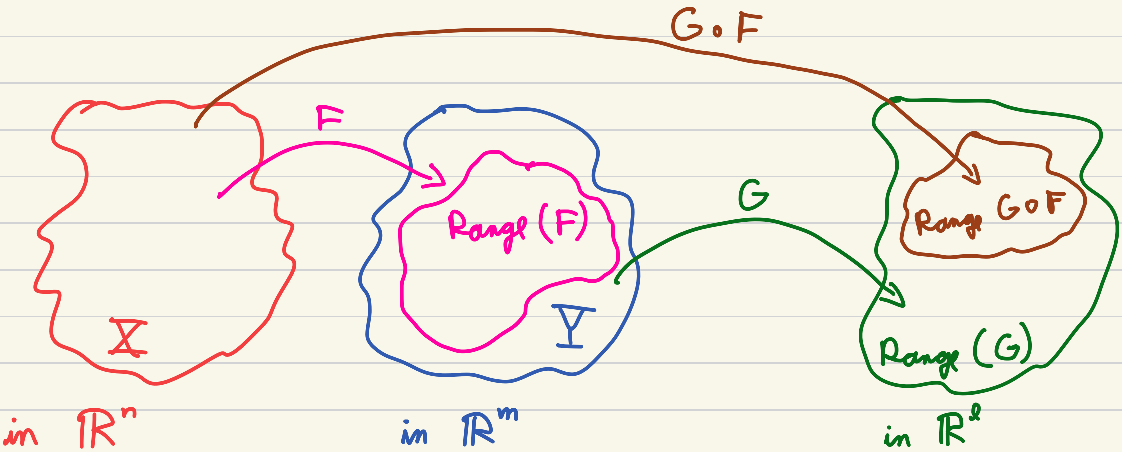
e.g. $\cdot\ f(x,y) = xy\,e^{y} + \sin(x)$ continuous on its domain $\mathbb{R}^2$

$\cdot\ F(x,y) = \left(x\cos(x),\ \frac{x}{y}\right)$ continuous on its domain

$$\text{domain}(F) = \mathbb{R}^2 \setminus \{(x,y) \in \mathbb{R}^2 : y = 0\}$$
$$\text{domain}(F) = \{(x,y) \in \mathbb{R}^2 : y \neq 0\}$$

Continuity of the composition

Theorem: If $F : X \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $G : Y \subseteq \mathbb{R}^m \to \mathbb{R}^l$ are continuous and satisfies that $\text{Range}(F) \subseteq Y$, then $G \circ F : X \subseteq \mathbb{R}^n \to \mathbb{R}^l$ is continuous.



e.g. $\cdot\ h(x,y) = e^{x^2 + \sin y} = g \circ f(x,y)$ this is continuous

$g(x) = e^x \quad g : \mathbb{R} \to \mathbb{R}$ is continuous

$f(x,y) = x^2 + \sin y \quad f : \mathbb{R}^2 \to \mathbb{R}$ continuous

$$\cdot\ h(x,y) = \underbrace{\cos(x^2 y)}_{h_1} + \underbrace{e^{\sin(x+y)}}_{h_2}$$

$f_1(x,y) = x^2 y$

$g_1(x) = \cos x$

$f_2(x,y) = x + y$

$g_2(x) = \sin x$

$\exp(x) = e^x$

$h_1(x,y) = g_1 \circ f_1 \qquad h_2(x,y) = \exp \circ g_2 \circ f_2$

$F(x,y) = (x^2 y, x+y)$

$G(x,y) = (x, \sin x)$

$H(x,y) = \cos x + e^y$

$H \circ G \circ F = h$

Recall: $f: X \subseteq \mathbb{R} \to \mathbb{R}$ is differentiable at $a \in X$ if $\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists, and we say $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

intuitively: $f'(a)$ is the rate of change of $f: X \subseteq \mathbb{R} \to \mathbb{R}$ as we move very close to a.

In Calc III we have many directions we can move along

Partial Derivatives

intuitively: rate of change as we move along the direction of one variable

def: For $f: X \subseteq \mathbb{R}^n \to \mathbb{R}$ and $\vec{x} = (x_1, \dots, x_n) \in X$ we say the partial derivative of f with respect to x_i (for some $i = 1, \dots, n$) is

$$\frac{\partial f}{\partial x_i} = \lim_{h\to 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(\vec{x})}{h}$$

we sometimes use other notation $f_x = \frac{\partial f}{\partial x}$ and $f_y = \frac{\partial f}{\partial y}$, for $f(x,y)$.

When we compute partial derivative with respect to x_i, we treat the rest of the variables as constants.

e.g. $f(x,y) = x^2 + y^2$

$$\frac{\partial f}{\partial x}(x,y) = 2x$$ we are treating y as constant

$$\frac{\partial f}{\partial y}(x,y) = 2y$$ we are treating x as constant

Important for partial derivatives: In practice $\frac{\partial f}{\partial x_i}$ is found by

1) Treat all variables other than x_i as constants

2) Take derivative as in Calc I using derivatives rules

3) If all points in the domain of f are in the domain of $\frac{\partial f}{\partial x_i}$, then you are done

4) For any point $\vec{a}$ in the domain of f but not in the domain of the $\frac{\partial f}{\partial x_i}$ you found, you may need to apply the partial derivative definition

e.g. $f(x,y) = e^{xy}\cos y$

$$\frac{\partial f}{\partial x}(x,y) = y e^{xy}\cos y$$

here y treated as constant

$$\frac{\partial f}{\partial y}(x,y) = x e^{xy}\cos y - e^{xy}\sin y$$

here x treated as constant

e.g. $f(x,y) = \frac{xy}{x^2+y^2}$

$$\frac{\partial f}{\partial x} = \frac{(x^2+y^2)y - xy2x}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial y}(x,y) = \frac{x(x^2+y^2) - xy2y}{(x^2+y^2)^2}$$

e.g. $f(x,y) = \begin{cases} 1 & \text{, if } (x,y)=0 \\ \frac{xy}{x^2+y^2} & \text{, if } (x,y)\neq 0 \end{cases}$

$$\frac{\partial f}{\partial x}(0,0) = 0 = \frac{\partial f}{\partial y}(0,0)$$

using the definition of part. derivative